| $x_{i+1}$ | $x_i$ | $x_{i-1}$ | | OP |
|---|---|---|---|---|
| 0 | 0 | 0 | | 0 |
| 0 | 0 | 1 | | 1 |
| 0 | 1 | 0 | | 1 |
| 0 | 1 | 1 | | 2 |
| 1 | 0 | 0 | | $\overline{2}$ |
| 1 | 0 | 1 | | $\overline{1}$ |
| 1 | 1 | 0 | | $\overline{1}$ |
| 1 | 1 | 1 | | 0 |

$0 \cdot 2^{i+1} + 0 \cdot 2^i = 0$

$0 \cdot 2^{i+1} + 1 \cdot 2^i = 1 \cdot 2^i$

$1 \cdot 2^{i+1} - 1 \cdot 2^i = 1 \cdot 2^i$

$1 \cdot 2^{i+1} + 0 \cdot 2^i = 1 \cdot 2^{i+1} = 2 \cdot 2^i$

$-1 \cdot 2^{i+1} + 0 \cdot 2^i = -2 \cdot 2^i$

$-1 \cdot 2^{i+1} + 1 \cdot 2^i = -1 \cdot 2^i$

$0 \cdot 2^{i+1} - 1 \cdot 2^i = -1 \cdot 2^i$

$0 \cdot 2^{i+1} + 0 \cdot 2^i = 0$

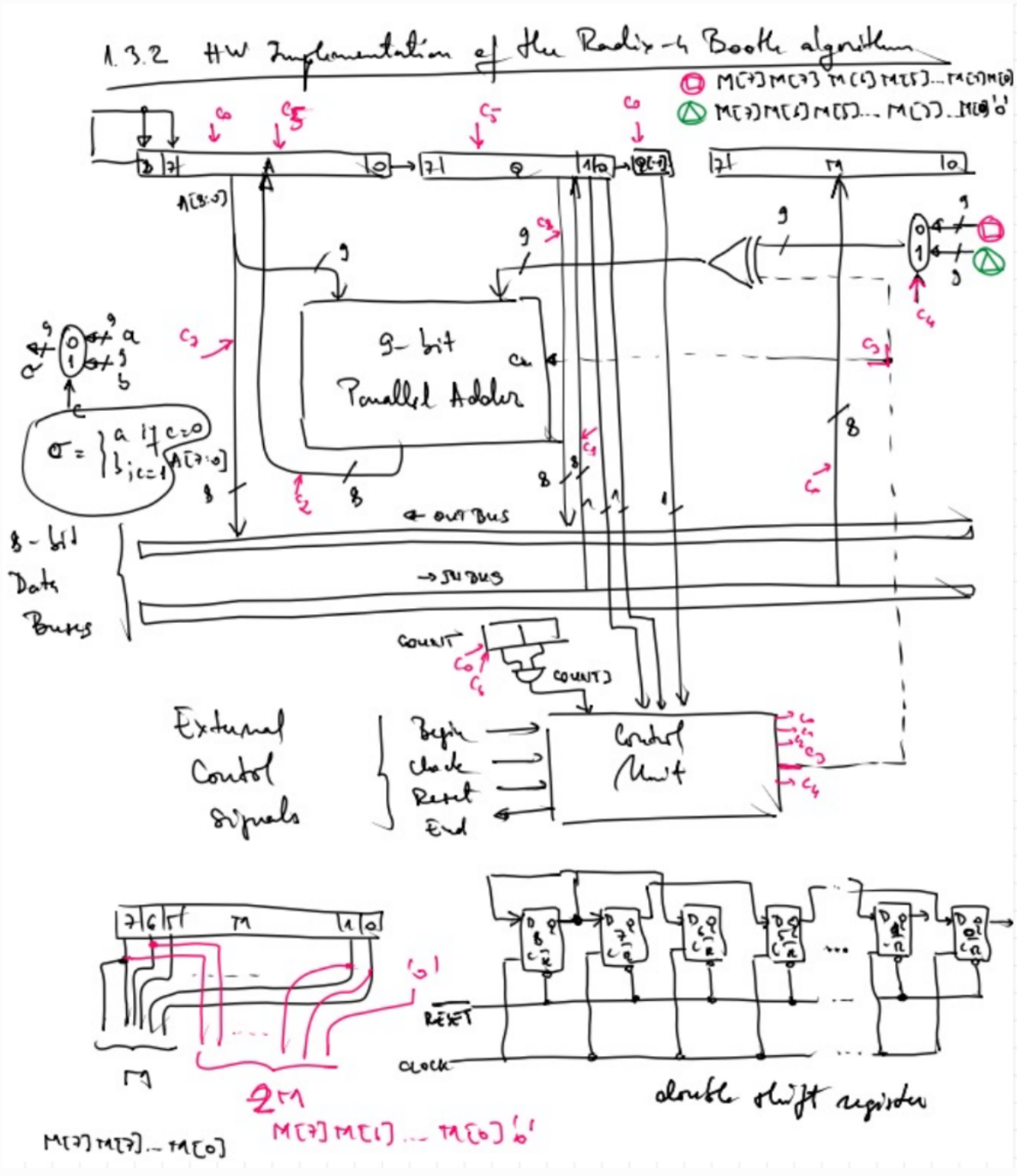
! Se înjumătățește timpul de execuție ← nr. de operații

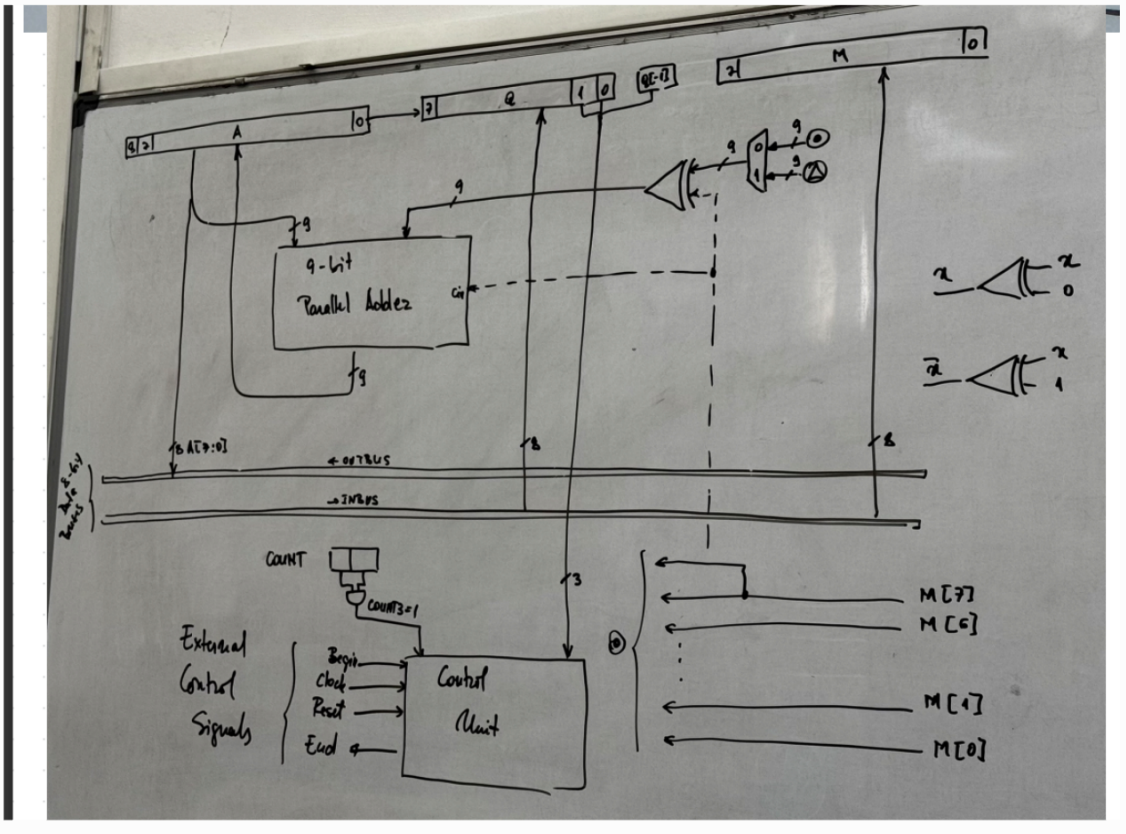
$CNT = \lceil \frac{n}{2} \rceil - 1$ CNT 3

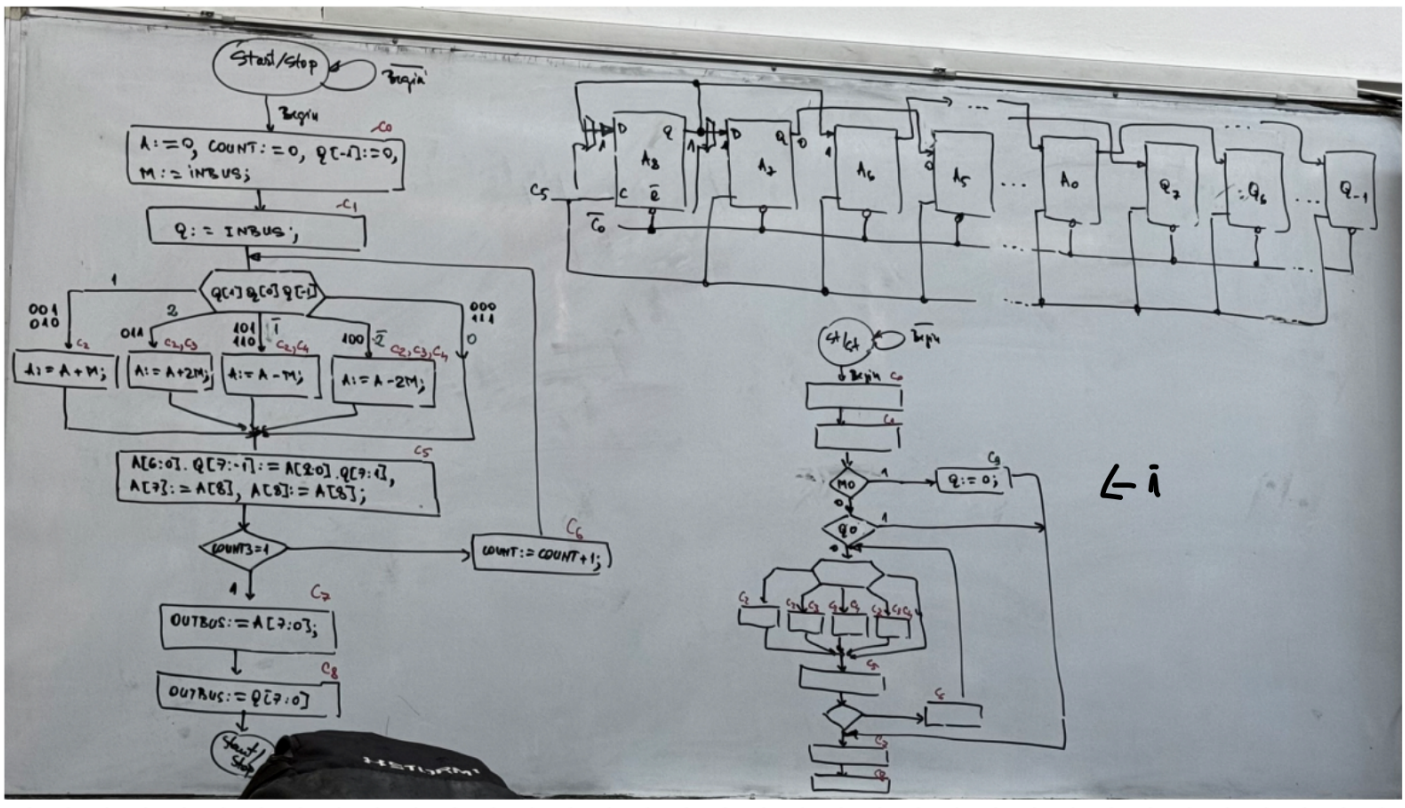
① $X = -123 = \underset{S}{1}\ 1\ 1\ 1\ \ 1\ 0\ 1\ 1_{SM} = 1\ 0\ 0\ 0\ \ 0\ 1\ 0\ 1_{C_2}$ → CNT 3

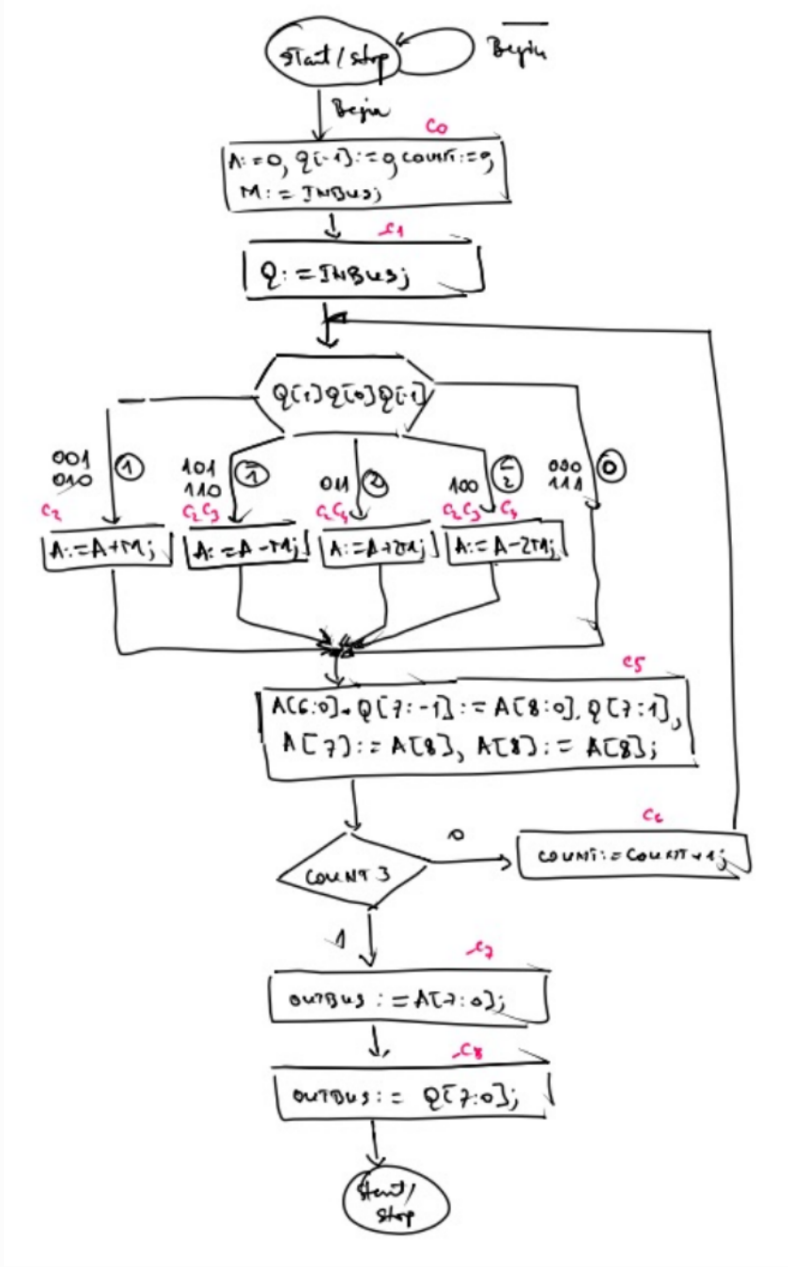
$Y = -71 = 1\ 1\ 0\ 0\ \ 0\ 1\ 1\ 1_{SM} = 1\ 0\ 1\ 1\ \ 1\ 0\ 0\ 1_{C_2}$

| CNT | A | Q | Q[-1] | M |
|---|---|---|---|---|
| 00 | 0 0000 0000 | 1011 1001 | 0 | 1 1000 0101 |
| ⊕ | 1 1000 0101 | | +M | -M |
| | 1 1000 0101 | | | 0 0111 1011 |
| | → RSHIFT cu 2 biți | | | 2M |
| | 1 1 1 1 0 0001 | 0 1 1 0 1 1 1 0 | 0 | 1 0000 1010 |
| 01 | ⊕ 0 1111 0110 | | -2M | -2M |
| c0 | 0 1101 0111 | | | 0 1111 0110 |
| 1 | → RSHIFT cu 2 | | | |
| | 0 0011 0101 | 1101 1011 | 1 | RSHIFT cu 2 |
| 10 | 0 0000 1101 | 0111 0110 | 1 | -M |
| 11 | ⊕ 0 0111 1011 | | | |
| | 0 1000 1000 | | | |
| | 0 0010 0010 | 0001 1101 | 1 | |

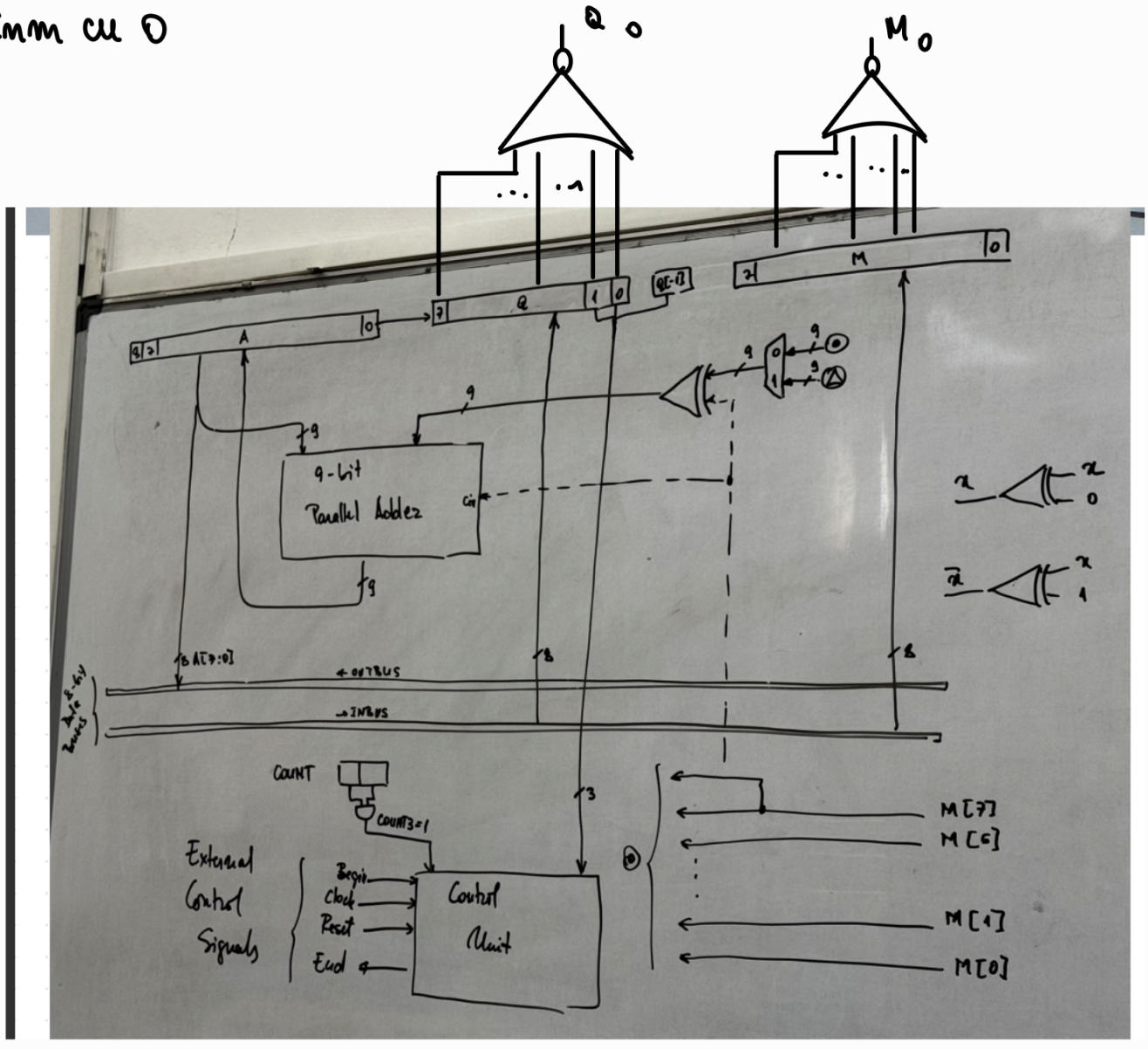

1.3.2 HW Implementation of the Radix-4 Booth algorithm
9-bit Parallel Adder
8-bit Data Buses
Count
Count=3
External Control Signals
Begin
Clock
Reset
End
Control Unit
RESET
CLOCK
double shift register

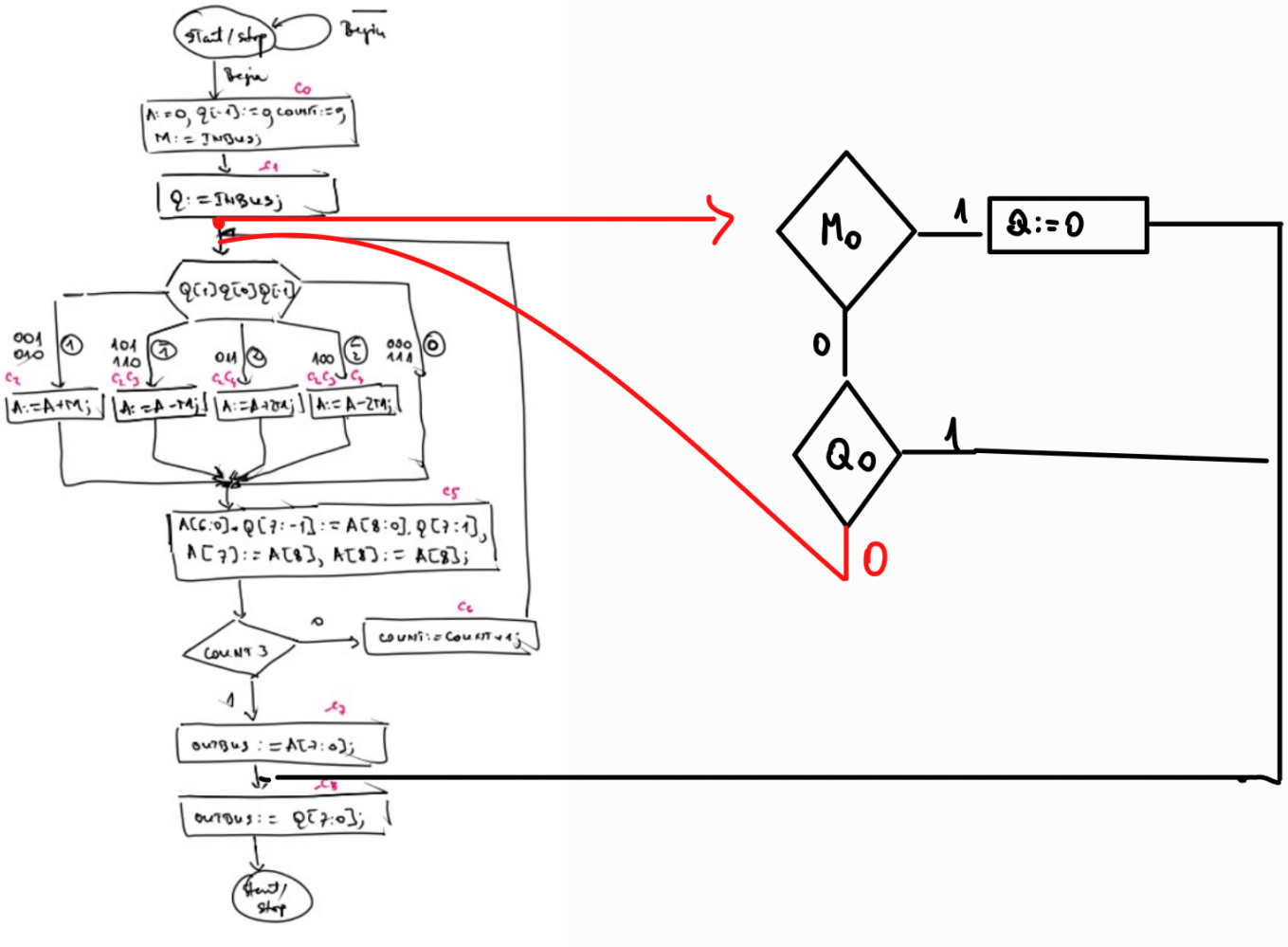

9-bit Parallel Adder
Count
Count=3
External Control Signals
Control Unit


Start/Stop
Begin
A:=0, COUNT:=0, Q[-1]:=0,
M:=INBUS;
Q:=INBUS;
Q[1]Q[0]Q[-1]
001
010
011
101
110
100
000
111
A:=A+M;
A:=A+2M;
A:=A-M;
A:=A-2M;
A[6:0], Q[7:-1]:=A[8:0], Q[7:1],
A[7]:=A[8], A[8]:=A[8];
COUNT3=1
COUNT:=COUNT+1;
OUTBUS:=A[7:0];
OUTBUS:=Q[7:0]
Start/Stop
Q0
MO
Q:=0;


Start / stop
Begin
Begin
A:=0, Q[-1]:=0, COUNT:=0,
M:=INBUS;
Q:=INBUS;
Q[1]Q[0]Q[-1]
001
010
101
110
011
100
000
111
A:=A+M;
A:=A-M;
A:=A+2M;
A:=A-2M;
A[6:0], Q[7:-1]:=A[8:0], Q[7:1],
A[7]:=A[8], A[8]:=A[8];
COUNT 3
COUNT:=COUNT+1;
OUTBUS:=A[7:0];
OUTBUS:=Q[7:0];
Start/
Stop

Imm cu 0


Q 0
M 0
A
Q
M
9-bit
Parallel Adder
BA[7:0]
←OUTBUS
→INBUS
COUNT
COUNT=1
External
Control
Signals
Begin
Clock
Reset
End
Control
Unit
M[7]
M[6]
M[1]
M[0]

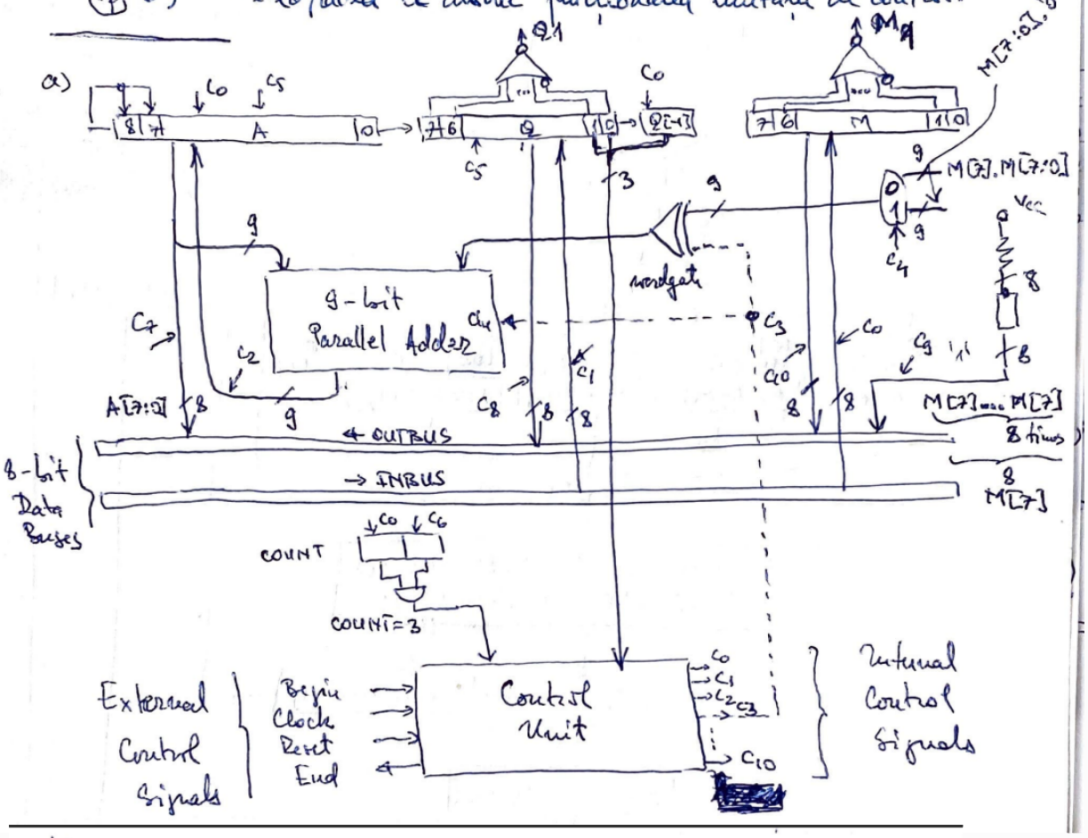

Start / stop
Begin
A:=0, Q[-1]:=0, count:=0;
M:=INBUS;
Q:=INBUS;
Q[1]Q[0]Q[-1]
A:=A+M;
A:=A-M;
A:=A+2M;
A:=A-2M;
A[6:0], Q[7:-1]:=A[8:0], Q[7:1],
A[7]:=A[8], A[8]:=A[8];
count=3
count:=count+1;
OUTBUS:=A[7:0];
OUTBUS:=Q[7:0];
Start / stop
M0
1
Q:=0
0
Q0
1
0

Rp ① Să se completeze descrierea unui înmulțitor Booth radix 4, astfel încât să trateze cazul special al înmulțirii cu 1 (unul dintre operanzi este egal cu 1). Se cer: | A1 B2

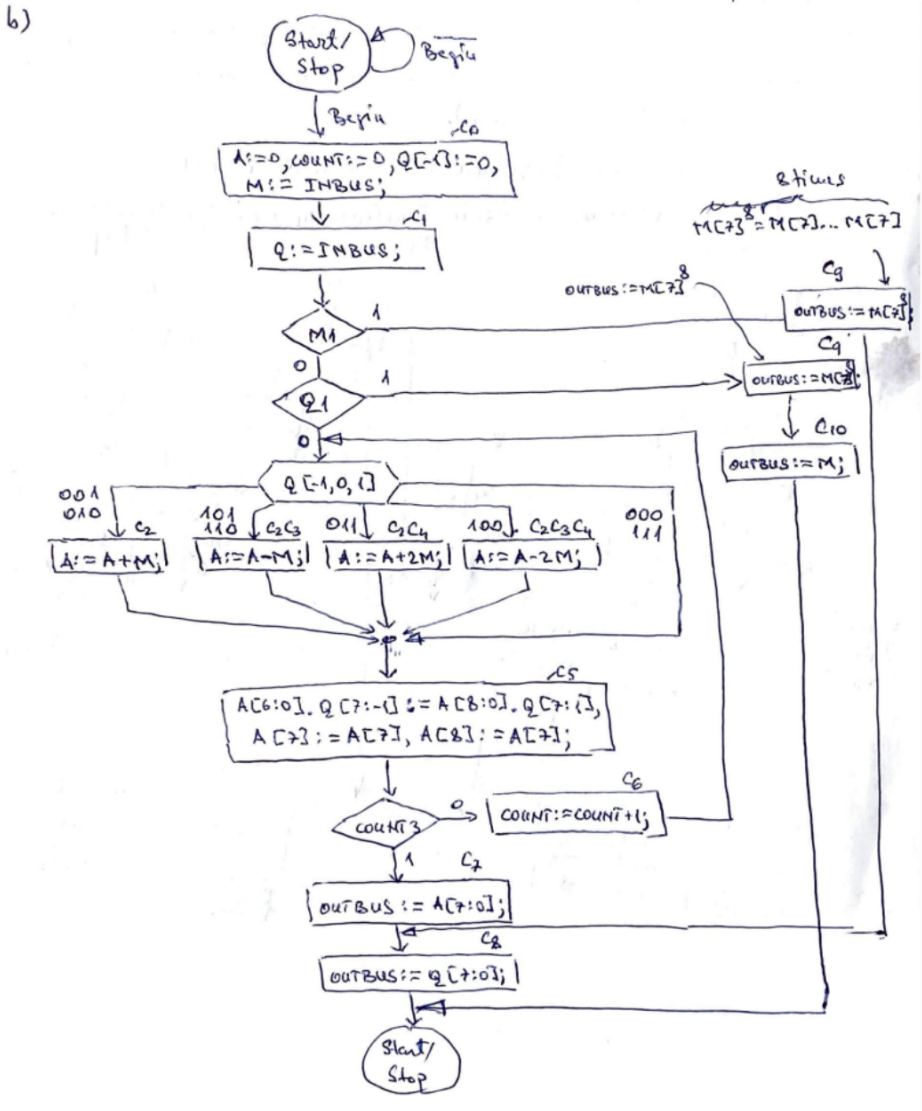
(1p) a) Schema bloc la nivel RTL + semnalele de control

(1p) b) Ordinograma ce descrie funcționarea unității de control.

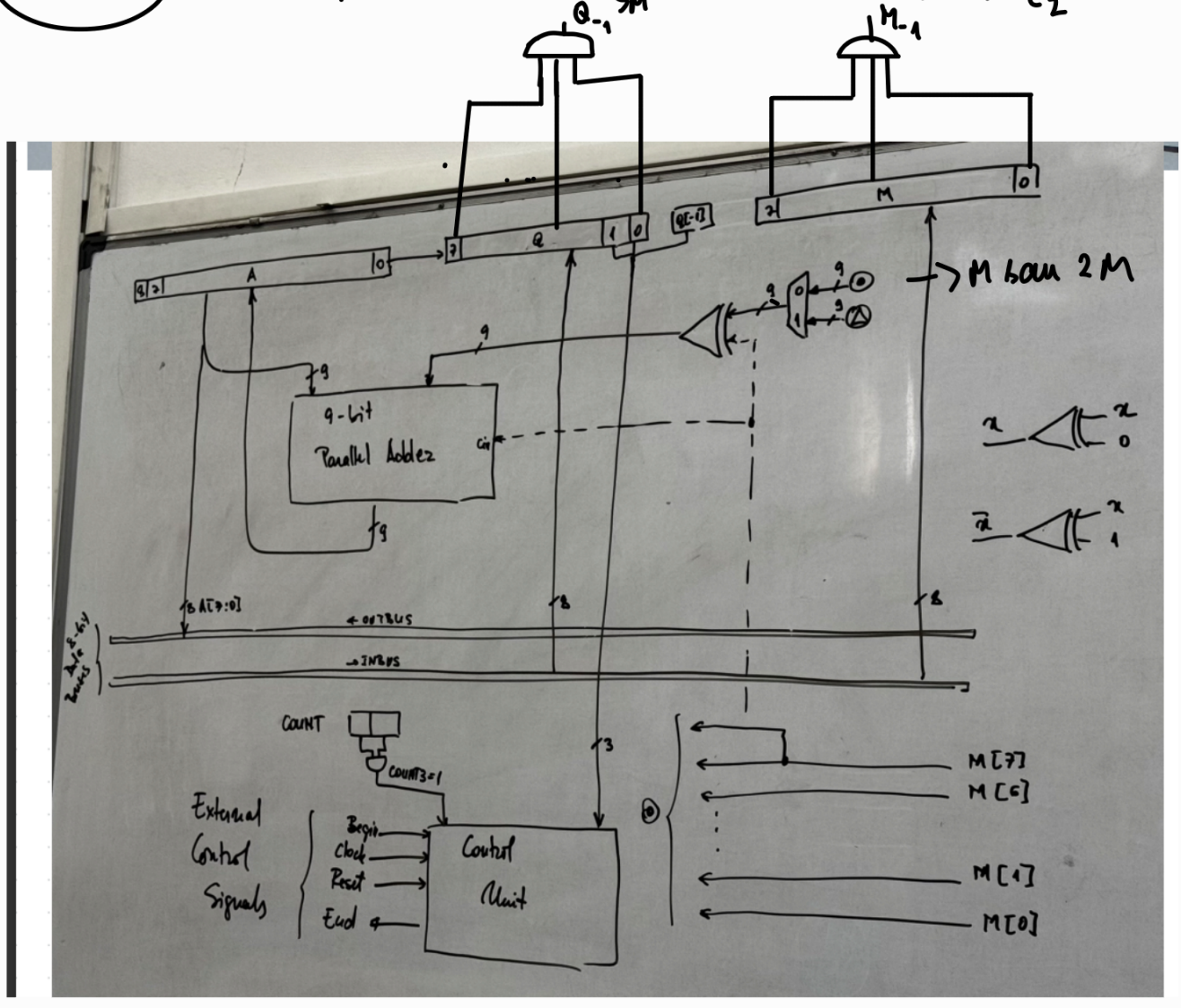
a)



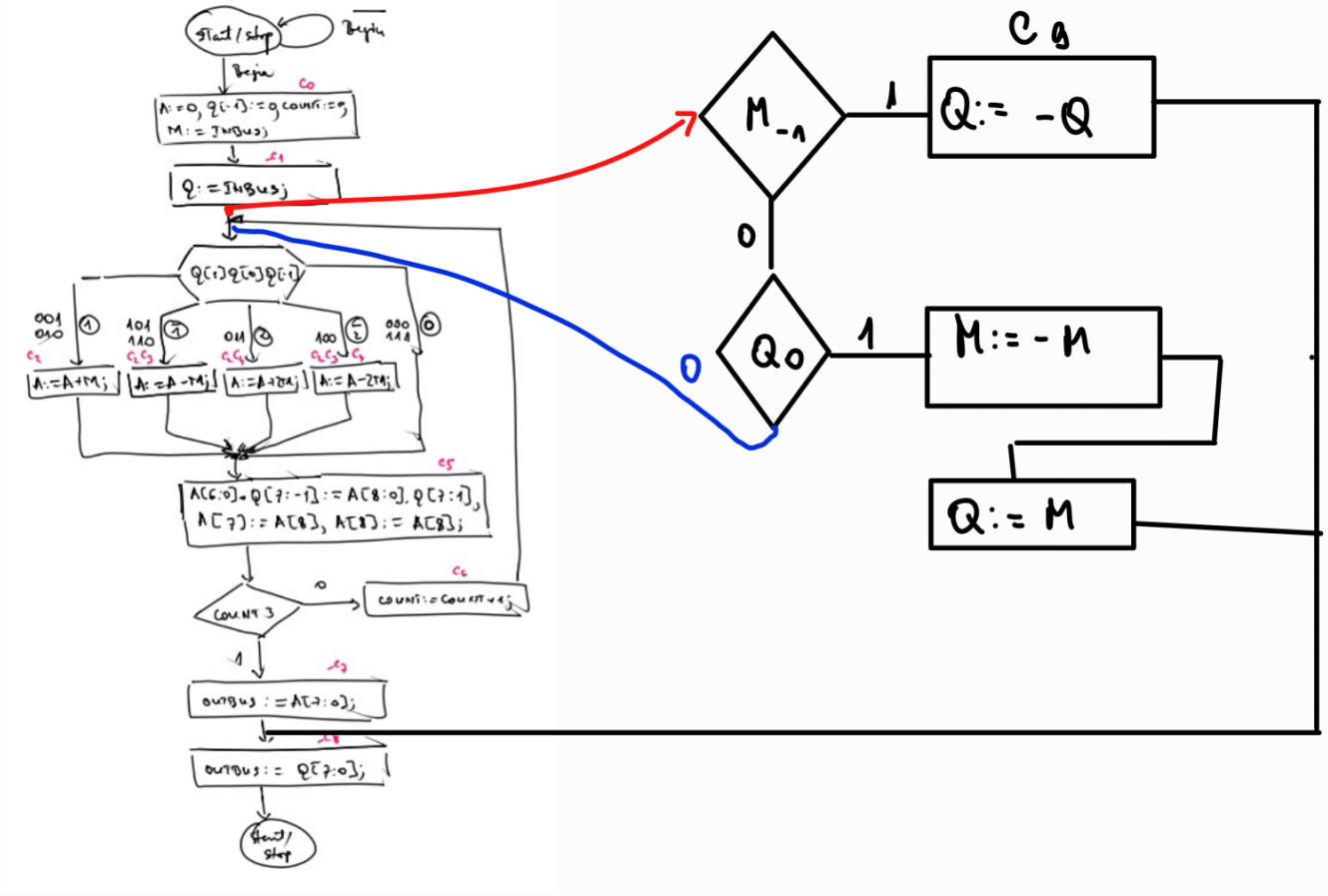

b)

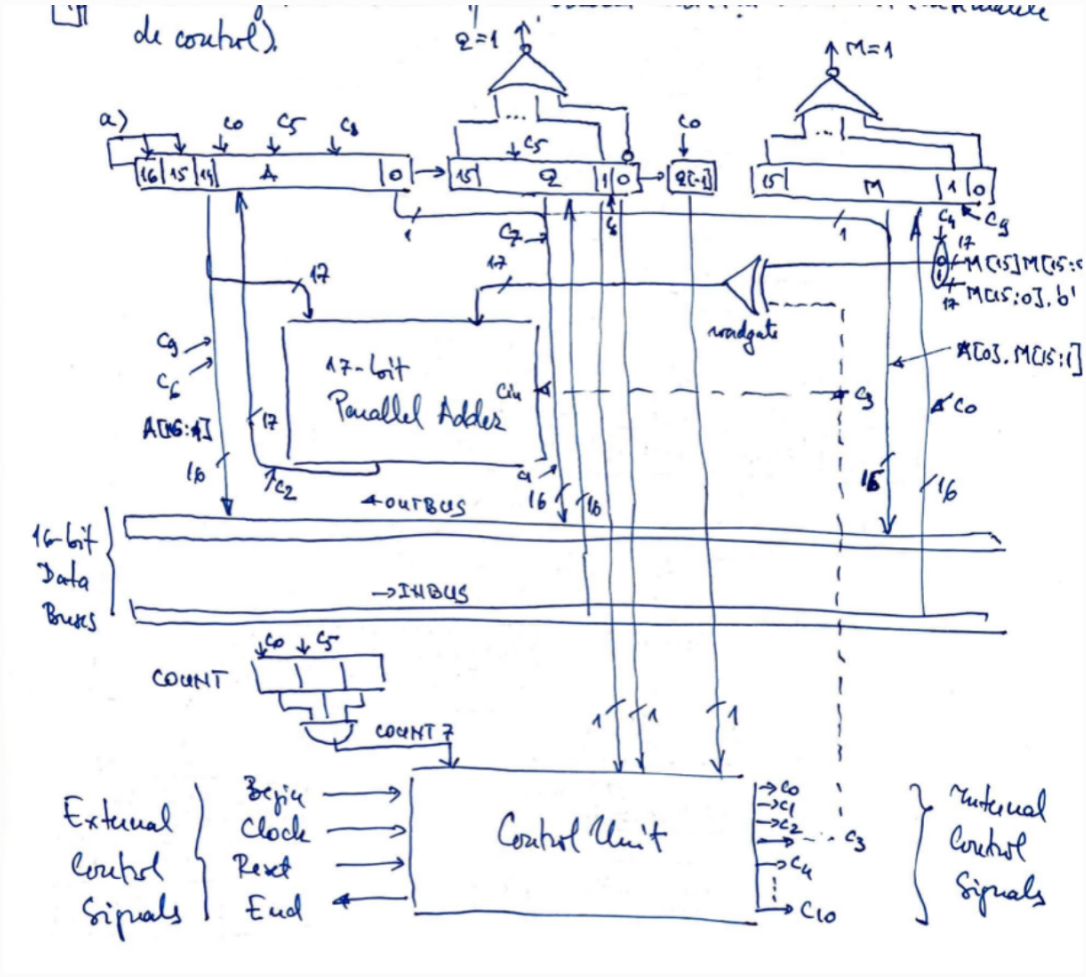


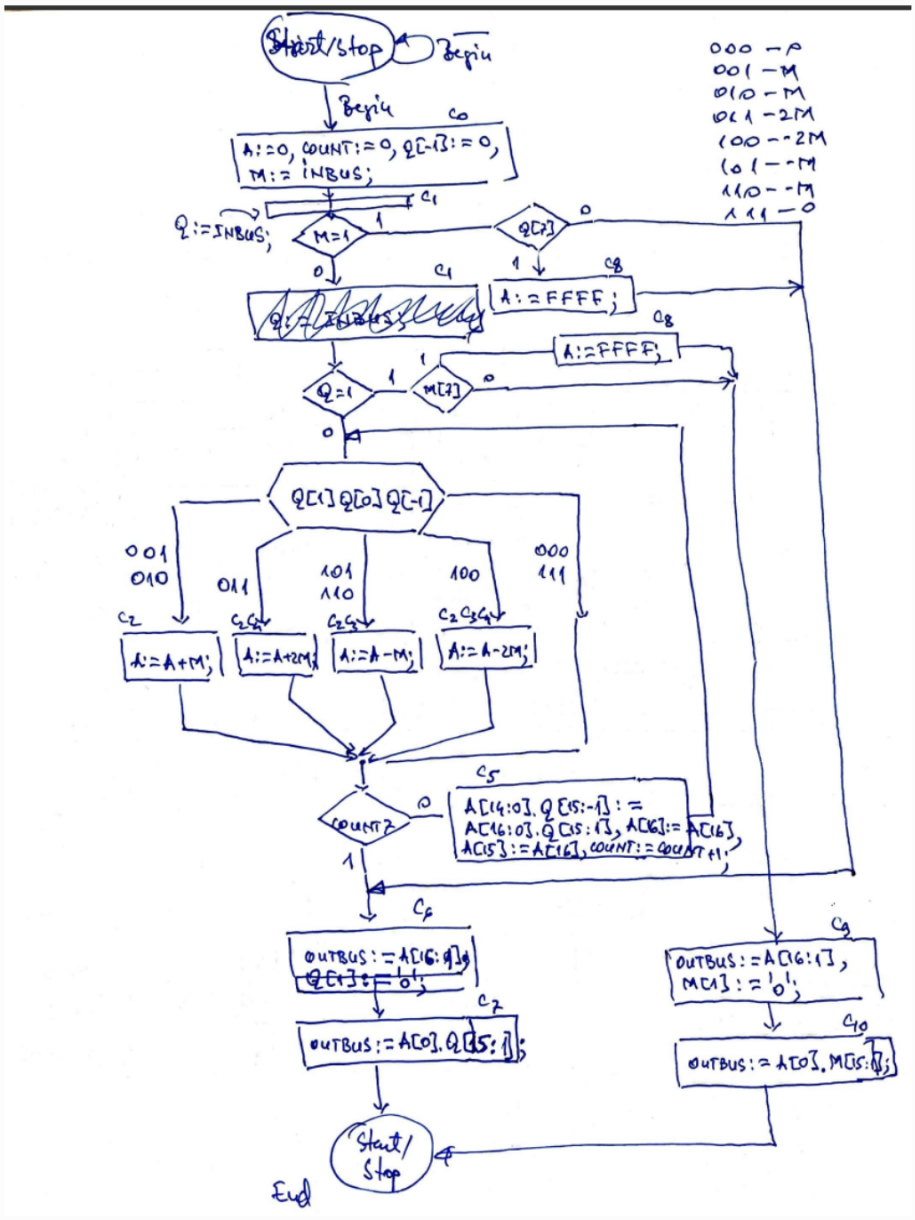
inm −1 = $1000\ 0001_{SM} = 1111\ 1111_{C_2}$




$Q_{-1}$ / $M_{-1}$ -> Multidgate XOR + $C_9/C_{10}$




17-bit Parallel Adder
OUTBUS
INBUS
16-bit Data Buses
COUNT
COUNT 7
Control Unit
External Control Signals
Internal Control Signals
Begin
Clock
Reset
End


Start/stop
Begin
A:=0, COUNT:=0, Q[-1]:=0, M:=INBUS;
Q:=INBUS;
M=1
Q[7]
A:=FFFF;
Q=1
M[7]
Q[1] Q[0] Q[-1]
A:=A+M;
A:=A+2M;
A:=A-M;
A:=A-2M;
COUNT 7
A[14:0], Q[15:-1] := A[16:0], Q[15:1], A[16]:=A[16], A[15]:=A[16], COUNT:=COUNT+1;
OUTBUS:=A[16:9], Q[1]:=0;
OUTBUS:=A[0], Q[15:1];
OUTBUS:=A[16:1], M[1]:='0';
OUTBUS:=A[0], M[15:1];
Start/ Stop
End